

Second Terminal Examination - 2079

Class : XI

Subject: Mathematics (0071)

Full Marks: 75

Time: 3 hrs.

SET 'A'

Pass Marks: 30

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all the questions.

Group 'A'

Rewrite the best alternative of each question in your answer sheet. [11 × 1 = 11]

- The converse of the statement $p \Rightarrow \sim q$ is
 a) $q \Rightarrow p$ b) $\sim q \Rightarrow p$ c) $q \Rightarrow \sim p$ d) $\sim p \Rightarrow q$
- For any two sets A and B , $B - A =$
 a) $A - B$ b) $\bar{A} \cap B$ c) $A \cap \bar{B}$ d) $\bar{A} \cup B$
- If $A = (-\infty, 1)$ and $B = [1, \infty)$ then $A \cap B$ is
 a) $\emptyset$ b) $\{1\}$ c) $\{0\}$ d) $(-\infty, \infty)$
- Let $A = \{1, 2\}$, then a relation on A defined by $x + y \geq 2$ is
 a) $\{\}$ b) $\{(1, 1)\}$
 c) $\{(1, 2), (2, 1), (2, 2)\}$ d) $A \times A$
- If A is a square matrix then $A + A^T$ is
 a) a symmetric matrix b) an identity matrix
 c) a skew symmetric matrix d) a triangular matrix

6. Which of the following matrix is not invertible?
 a) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ b) $\begin{bmatrix} 5 & 3 \\ 1 & 7 \end{bmatrix}$ c) $\begin{bmatrix} -2 & -4 \\ 3 & 6 \end{bmatrix}$ d) $\begin{bmatrix} 4 & 5 \\ 1 & 2 \end{bmatrix}$
7. The value of $\tan(\tan^{-1}x + \cot^{-1}x)$ is
 a) $\frac{\pi}{2}$ b) 0 c) 1 d) undefined
8. Lines $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$ are parallel if
 a) $\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$ b) $\frac{a_1}{a_2} \neq \frac{b_1}{b_2} = \frac{c_1}{c_2}$
 c) $\frac{a_1}{a_2} = \frac{b_1}{b_2}$ d) $\frac{a_1}{a_2} \times \frac{b_1}{b_2} = -1$.
9. Which points lie on the same side with respect to the line $x - y = 3$?
 a) $(3, -2)$ & $(4, 0)$ b) $(3, 0)$ & $(2, 3)$
 c) $(3, 0)$ & $(0, -3)$ d) $(2, 3)$ & $(2, -4)$
10. The value of $\lim_{x \rightarrow 0} \frac{\tan^{-1}x}{x}$ is
 a) 0 b) 1 c) $\frac{1}{3}$ d) 3
11. If $f(x) = \frac{1}{x}$, then $f'(x)$ at a point $(1, 1)$ is
 a) $-\frac{1}{x^2}$ b) -1 c) 1 d) not exist

Group 'B'

Short Answer Questions.

[8 × 5 = 40]

12. a) Define contradiction. Prove that $p \wedge (\sim p)$ is a contradiction. [2]
 b) If A , B and C are the subsets of universal set U , prove that
 $A - (B \cap C) = (A - B) \cup (A - C)$. [3]

13. a) A function $f: \mathbb{R} \rightarrow \mathbb{R}$ is defined by $f(x) = 2x - 5, x \in \mathbb{R}$.
Show that f is bijective. Also, obtain the formula for f^{-1} . [3]
- b) Find the domain and range of a function $f(x) = x^2 + 4x - 3$. [2]
14. a) If $A = \begin{bmatrix} 2 & 1 \\ 5 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & -1 \\ -5 & 2 \end{bmatrix}$ then verify $AB = BA = I$.
Also comment on the result. [2]
- b) Prove that : $\begin{vmatrix} x+y & x & z \\ y+z & y & y \\ z+x & z & x \end{vmatrix} = (x+y+z)(x-z)(y-z)$ [3]
15. a) Solve for the general solution of x .
 $\cos x + \sqrt{3} \sin x = \sqrt{2}$ [2]
- b) Solve for x : $\tan^{-1} 2x + \tan^{-1} 3x = \frac{\pi}{4}$ [3]
16. a) Find the distance between parallel lines $2x + 3y - 7 = 0$ and
 $4x + 6y + 9 = 0$. [2]
- b) If p is the length of perpendicular dropped from the origin on the line
 $\frac{x}{m} + \frac{y}{n} = 1$, prove that $\frac{1}{m^2} + \frac{1}{n^2} = \frac{1}{p^2}$. [3]
17. If P_1 and P_2 be the length of the perpendiculars drawn from the origin upon the straight lines whose equation are $x \sec \theta + y \operatorname{cosec} \theta = a$ and
 $x \cos \theta - y \sin \theta = a \cos 2\theta$, prove that $4P_1^2 + P_2^2 = a^2$. [5]
18. a) Evaluate: $\lim_{x \rightarrow 0} \frac{1 - \cos 3x}{x^2}$ [2]
- b) Define continuity of a function at a point. Find the value of q for which
the function $f(x) = \begin{cases} \frac{2x^2 - 18}{x - 3} & \text{for } x \neq 3 \\ q & \text{for } x = 3 \end{cases}$ is continuous at $x = 3$. [3]

19. a) Find from first principles, the derivative of e^x . [2]
 b) Find $\frac{dy}{dx}$ when $y = \frac{1}{\sqrt[3]{3x^2-4x-1}}$ [3]

Group 'C'

Long Answer Questions.

[3 × 8 = 24]

20. a) Construct the truth table for the compound statement $(\sim p) \Rightarrow q$. [2]
 b) If $A \cap B = \phi$, prove that $A \cup \bar{B} = \bar{B}$. [2]
 c) For any real number 'x' and any positive real number 'a' prove that $|x| \leq a \Rightarrow -a \leq x \leq a$. Using the implication, rewrite the inequality $|4x + 2| \leq 6$ without using absolute value sign and solve for x. [2+2]
21. a) Let $A = \{-1, 0, 1, 2, 3\}$, $B = \left\{-2, 0, -\frac{1}{2}, \frac{1}{4}, \frac{2}{5}, 1\right\}$ and a function $f: A \rightarrow B$ be defined by $f(x) = \frac{x-1}{x+2}$. Find the range of f . Is the function f is one to one and onto both? If not, how can you make it one to one and onto both? [2+1+1]
 b) Evaluate: $\lim_{x \rightarrow 0} \frac{(a+x) \sec(a+x) - a \sec a}{x}$ [4]
22. a) Find from first principles, the derivative of $\sqrt{3x-2}$. [4]
 b) Find $\frac{dy}{dx}$ if $x^2 y^2 = x^2 + y^2$ [4]

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